

Empirical Exercise in Value at Risk

Assume that you are the risk manager of a retirement fund. The fund has €500 million in cash to invest in fixed-income markets. You decide to allocate this amount equally across 50 bond ETFs of your choice. Your objective is to estimate the market risk of this portfolio using alternative Value-at-Risk methods and to determine which method provides the best performance.

Data Requirement

Choose 50 bond ETFs and form an equally weighted portfolio. Download two years of daily historical data for all selected ETFs. In order to compute returns, use adjusted closing prices whenever available. Assume that the regulator requires the computation of the 90% VaR.

Question A

Estimate the next-day VaR of the portfolio at the 90% confidence level using the following methods:

1. Historical simulation.
2. Parametric VaR under the assumption of normally distributed returns.
3. Parametric VaR under the assumption of Student- t distributed returns.

You must write the corresponding Python code for all methods and include it in an appendix to your essay.

Question B

Use backtesting to evaluate the forecasting performance of the three VaR methods in Question A and determine which method should be preferred.

Question C

Estimate the next-week VaR (5 trading days) at the 90% confidence level using:

1. Historical simulation.
2. Parametric VaR under the assumption of normally distributed returns.
3. Parametric VaR under the assumption of Student- t distributed returns.

Submission Instructions

Submit your work as one PDF file. The essay should contain a description of the data, the methodology, the empirical results, the backtesting evidence, the final recommendation, and an appendix with all Python code.